

20CS3T04 – Database Management Systems

UNIT 1: Introduction to DBMS

Introduction

A Database Management System (DBMS) is software that manages databases and provides mechanisms for storing, retrieving, and managing data.

This unit introduces DBMS concepts and architecture.

Detailed Explanation

Database Concepts

A database is a collection of related data. DBMS ensures data integrity and security.

Advantages of DBMS

It reduces redundancy and improves data consistency.

Applications

DBMS is used in banking, education, and e-commerce systems.

Summary

This unit introduces database systems and their importance.

UNIT 2: Relational Model

Introduction

The relational model organizes data into tables with rows and columns.

This unit focuses on relational databases.

Detailed Explanation

Tables and Keys

Tables consist of rows and columns, with primary keys uniquely identifying records.

Relationships

Relationships define how tables are connected.

Applications

Relational databases are widely used in modern applications.

Summary

This unit explains relational data organization.

UNIT 3: SQL and Database Design

Introduction

SQL is used to interact with databases. Database design ensures efficient data storage.

Detailed Explanation

SQL Commands

Commands such as SELECT, INSERT, UPDATE, and DELETE are used.

Normalization

Normalization reduces redundancy and improves consistency.

Applications

SQL is used in data management and application development.

Summary

This unit covers SQL and design techniques.

UNIT 4: Transaction Management

Introduction

Transactions ensure reliable database operations.

Detailed Explanation

ACID Properties

Transactions follow Atomicity, Consistency, Isolation, and Durability.

Concurrency Control

Ensures multiple transactions execute safely.

Applications

Transaction management is used in banking systems.

Summary

This unit explains reliable data operations.

UNIT 5: Indexing and Query Processing

Introduction

Indexing improves database performance, and query processing retrieves data efficiently.

Detailed Explanation

Indexing Techniques

Indexes speed up data retrieval.

Query Processing

DBMS optimizes queries for efficient execution.

Applications

Used in search engines and large databases.

Summary

This unit introduces indexing and query optimization.